2023 Joint European Conference on Networks and Communications & 6G Summit (EuCNC/6G Summit) | 979-8-3503-1102-0/23/$31.00 ©2023 IEEE | DOI: 10.1109/EUCNC/6GSUMMIT58263.2023.10188298

Simulation of Data Hijacking Attacks for a 5G-Advanced Core Network

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***Abstract*—The 5G mobile communication technology provides a faster transmission speed, larger bandwidth, and the ability to connect a greater number of devices than 4G. However, ensuring the successful transition to 5G-Advanced requires addressing various security vulnerabilities and threats. It is imperative in 5G-Advanced and higher mobile communications to address the security risks that have arisen in current mobile communication systems. In this paper, we perform a simulation of scenario-based data hijacking attacks for a 5G-Advanced core network. The conducted simulation results demonstrate that two different data hijacking attacks are possible, with sensitive information being vulnerable to exploitation through security weaknesses such as the lack of encryption for internal communication and inadequate authentication of internal components in the 5G-Advanced core network.**

Keywords—5G-Advanced; Attack Simulation; Data Hijacking; Network Security

I. Introduction

The wireless communication industry is experiencing rapid growth and development. Advances in telecommunications technology have led to improved infrastructure and exponential growth. In 2019, three major South Korean mobile carriers launched the first 5G mobile communication service, and by 2025, 5G is expected to have significant coverage [1]. Experts predict that 6G mobile communication, the next-generation network, will become widely available as 5G commercialization continues [2]. Edge Intelligence, THz Communication, MEC, and Network Slicing are being explored as key technologies for the next generation of mobile communication, but these concepts have yet to be fully integrated into 5G networks, hindering some applications from meeting the requirements for high-speed data transfer and real-time computing. The commercialization of 5G advancements has facilitated the integration of the Internet of Things (IoT), connecting a vast number of devices and cyber-physical systems as it moves into the 5G-Advanced era [3]. However, the growth of new service types and the exponential increase in connected devices have also led to an increase in security threats.

The successful implementation of 5G-Advanced requires addressing a variety of potential security vulnerabilities and issues. Previous networks have faced problems such as user privacy breaches, wireless interface threats, and inadequate authentication procedures between the User Equipment (UE) and the network [4]. These vulnerabilities and issues highlight that 5G systems remain at risk from active attacks and wireless interface threats. It is essential to address the security risks that have arisen in previous mobile communication systems as we transition to 5G-Advanced and beyond.

To successfully transition to 5G-Advanced and 6G, it is necessary to address both the security threats that are expected to emerge in future mobile communication systems as well as the security threats that have already occurred in previous mobile communication systems. With the advent of 5G-Advanced and 6G, the concept of openness has gained importance, and the trend is to privatize the core networks, edge, and RAN using open platforms. However, it should be noted that the privatization of mobile communication networks may lead to reduced reliability of the network for lawful users. To ensure the security of 5G-Advanced and 6G networks, it is crucial to proactively identify and address potential attacks within the core network that may not have been previously considered.

In this paper, we outline our experimental approach to 5G-Advanced by setting up a simulation testbed. We carry out two different data hijacking attacks on the testbed to demonstrate the vulnerability of sensitive information to exploitation. The simulation results show that these types of attacks are possible due to security weaknesses such as the absence of encryption for internal communication and insufficient authentication of internal components in the core network for 5G-Advanced.

The paper is organized as follows: In Section II, we describe the structure of our testbed and the UE registration. In Section III, we present the results of our simulation-based data hijacking attacks, including a comprehensive analysis. The limitations of our study are discussed in Section IV and the paper concludes in Section V.

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II. Preliminaries

A. Simulation Testbed

The objective of the 5G system is to provide connectivity to the UE. The 5G system registers the UE in the network through a procedure in the Control Plane (CP) and establishes a User Plane (UP) path to the data network. This procedure involves communication between several components, from the network edge to the core, and is facilitated by the standard architecture defined in 3GPP [5].

The 5G core uses Network Function Virtualization (NFV) and Software Defined Networking (SDN) technologies to manage the CP and UP of the UE through Network Functions (NFs). The next-generation Radio Access Network (RAN) sets up the wireless network between the UE and the base station (i.e., gNB) and facilitates communication between the gNB and the 5G core through the Next Generation Application Protocol (NGAP) interface.

In this paper, we use Free5GC and UERANSIM to build our testbed shown in Fig. 1. Free5GC is an open-source 5G core network that implements 3GPP Release 15 and later architectures [6]. UERANSIM is an open-source tool that simulates the RAN in conjunction with the core network by configuring the gNB and the UE [7].

All NFs of Free5GC and UERANSIM are set up using docker containers to ensure efficient operation. After the configuration is completed, Free5GC and UERANSIM are linked through an interface to form the 5G network, which is capable of utilizing a data network.

B. UE Initial Registration Process

In the 5G network, the initial UE is ultimately registered through a process that involves authentication [8], [9]. This results in the establishment of a successful Protocol Data Unit (PDU) session between the UE and the User Plane Function (UPF), allowing the UE to access and utilize the data network. The procedure for registering the UE is depicted in Fig. 2. The description of each step is as follows:

Steps (1) to (4) describe the process of preparing NGAP and Non Access Stratum (NAS) connections for the UE initial registration. Through this process, the Access and Mobility Management Function (AMF) confirmed that the UE made the initial registration request. (1) The gNB requests a NGAP Setup from the AMF. Upon receiving a response, it establishes a connection between the gNB and the AMF; (2) The UE

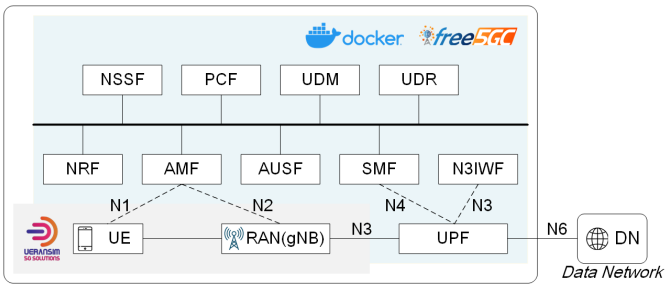


Fig. 1. Proposed 5G-Advanced Testbed Architecture

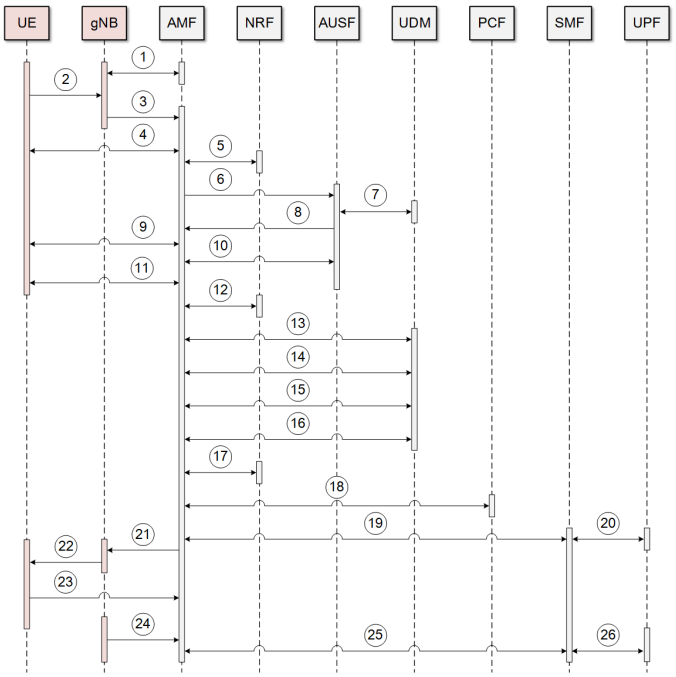


Fig. 2. 5G UE Initial Registration Process

requests the gNB to register the UE; (3) The gNB requests the AMF initial UE message for the 5G UE registration; (4) The AMF requests the UE for its downlink Non Access Stratum (NAS) identity and receives an uplink NAS identity response.

Steps (5) to (11) describe the process of authenticating the UE that has requested the UE initial registration. Through this process, the UE is authenticated to the 5G network, and a key agreement is made. (5) The AMF requests and receives the Authentication Server Function (AUSF) Discover message from the Network Repository Function (NRF) in order to perform authentication of the UE; (6) The AMF requests the UE authentication information to the AUSF; (7) The AUSF requests the UE authentication information to the User Data Management (UDM) and receives a response; (8) The AMF receives the UE authentication information from the AUSF; (9) The AMF requests the UE for its downlink NAS authentication and receives an uplink NAS authentication response; (10) The AMF requests and receives the confirmation from the AUSF for 5G authentication and a key agreement; (11) The AMF requests the UE for its downlink NAS security mode and receives an uplink NAS security mode response.

Steps (12) to (16) describe the process of obtaining the subscription information for the UE. Through this process, the AMF obtains the subscription information of the UE and grants authorization as appropriate. (12) The AMF requests and receives a UDM Discover message from the NRF; (13) The AMF requests the UE slice selection information from the UDM and receives a response; (14) The AMF requests the Access and Mobility Management (AM) Subscription information from the UDM and receives a response; (15) The AMF requests the Session Management (SM) Subscription information from the UDM and receives a response; (16) The AMF requests and receives a Subscriber Data Management (SDM) Subscription from the UDM.

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Steps (17) to (20) are the process of establishing a session after controlling the policy. Through this process, a PDU session and a Packet Forwarding Control Protocol (PFCP) session are created. (17) The AMF requests and receives a Policy Control Function (PCF) Discover message from the NRF; (18) The AMF requests and receives a response from the PCF to perform the UE's Access and Mobility (AM) policy control; (19) The AMF requests and receives a response to create a SM Context to configure a PDU session through the Session Management Function (SMF); (20) The SMF requests and responds to a Packet Forwarding Control Protocol (PFCP) session establishment in order to perform the signaling of CP and UP from the UPF.

Steps (21) to (24) describe a process of notifying the UE that the initial registration has been completed. Through this process, the UE successfully registers with the 5G network by completing NGAP and NAS registration. (21) The AMF requests an NGAP Initial Context Setup from the gNB, indicating that the UE registration has been accepted; (22) The gNB transmits a message to the UE to indicate that the NAS registration is accepted; (23) The UE transmits a message to the AMF to indicate that the NAS registration is complete; (24) The AMF responds to the gNB with the success message of the Initial Context.

Steps (25) to (26) are a process of updating UE information. Session information is modified through this process. (25) The AMF requests and receives a PDU session SM Context update from the SMF; (26) The SMF requests and receives a PFCP session modification from the UPF.

III. Data Hijacking Attack Simulation

In this section, the attack environment is introduced, simulating of data hijacking attacks. The demonstration of these attacks is then carried out in two distinct scenarios.

A. Experiment Environment

The simulation environment for data hijacking attacks has been set up by configuring the NFs of Free5GC as containers using Docker to form a 5G-Advanced core network. UERANSIM is also configured as a container to form the NG-RAN and is connected to Free5GC to simplify the simulation environment. Docker is an open-source container runtime that makes it simple to run containers and is appropriate for container-based simulation environments. To execute the attacks, an additional container named attack_node has been added. The attack_node was built via Docker, which used the standard Ubuntu 20.04

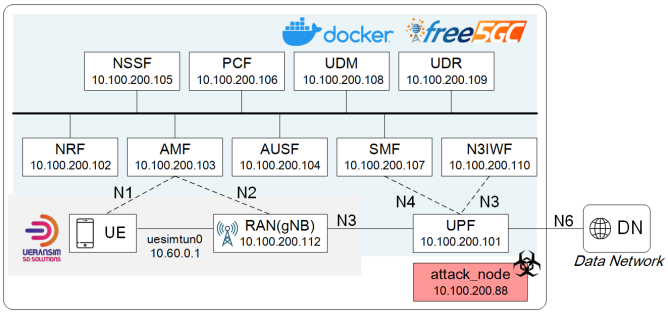


Fig. 3. Testbed Architecture

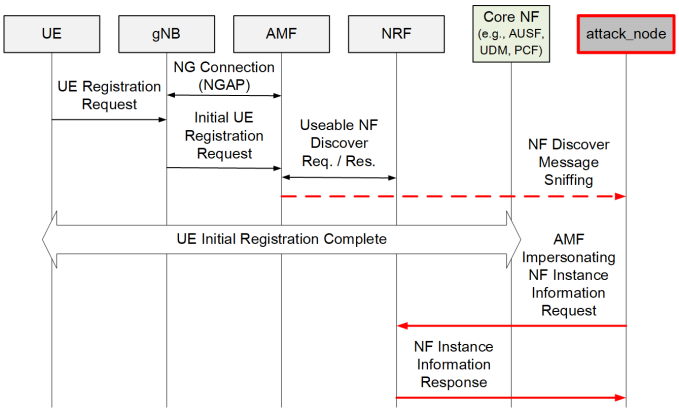


Fig. 4. NF Data Hijacking Attack Scenario

image downloaded from the Docker Hub Image Library and provided a field to carry out the attack [10]. It is connected to the 5G-Advanced network bridge and equipped with a packet analyzer, Wireshark. As depicted in Fig. 3, each NF is assigned fixed IP addresses to facilitate our experiment.

B. NF Data Hijacking Attack

The NF data hijacking attack is an attack that exploits vulnerabilities in the network by impersonating the AMF on an attack_node and hijacking NF data from the NRF. This scenario is depicted in Fig. 4.

1) Attack Scenario: The AMF initiates a request to the NRF during the initial registration process of the UE to the core network to discover the relevant NFs, such as the AUSF and PCF, in order to gather information about the 5G Core NF. An attack_node connected to the network, on the other hand, can intercept this process by sniffing the NF Discovery message. Once the UE registration process is complete, the attack_node uses the information obtained from the sniffed message to send an NF instance information request while impersonating the AMF. Without performing proper authentication procedures, the NRF responds with the requested NF instance information.

2) Attack Simulation: Fig. 5 depicts the sniffed NF Discover message, which was transmitted by the AMF to the NRF. After the completion of the initial UE registration, the attack_node modifies the sniffed message and sends it to the NRF, impersonating the AMF, through a CURL command-based HTTP GET request. The used command is as follows:

```
$ docker exec attack_node curl "http://10.
100.200.102:8000/nnrf-disc/v1/nf-instances
?requester-nf-type=AMF&target-nf-type=
[NF name]"
```

The `docker exec` command is used to execute the curl command within the attack_node container. The attack_node impersonates the AMF by sending a request to the NRF with `request-nf-type=AMF` and requests information about the NF by specifying `target-nf-type=[NF name]`. The

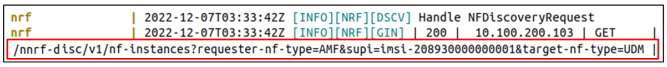


Fig. 5. Sniffed NF Discover Message

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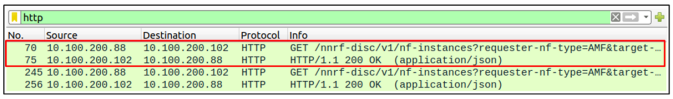


Fig. 6. NF Data Hijacking Attack Packet

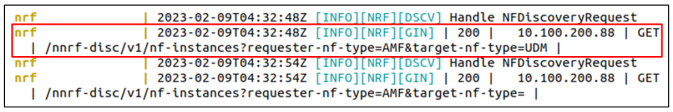


Fig. 7. NF Data Hijacking Attack Log



Fig. 8. NF Data Hijacking Attack Result

`[NF name]` field can be filled with the name of the NF for which information is required, and if left blank, all the information about the NFs stored in the NRF will be requested. Figs. 6 and 7 depict the attack packets and the logs produced by the NRF as a result of the attack.

Fig. 8 depicts the outcome of the attack. This information includes the API version of the NF, Mobile Country Codes (MCC) information, Mobile Network Codes (MNC) information, IP information, IP endpoint, service instance, and service name. The obtained information can be used to launch secondary attacks such as a Denial of Service (DoS) attack or a replay attack.

C. UDM User Data Hijacking Attack

The UDM user data hijacking attack is an attack intercepting the Subscription Permanent Identifier (SUPI) information of an UE at an attack_node. This information is then used to gain UE's data that is registered in the UDM. This scenario is depicted in Fig. 9

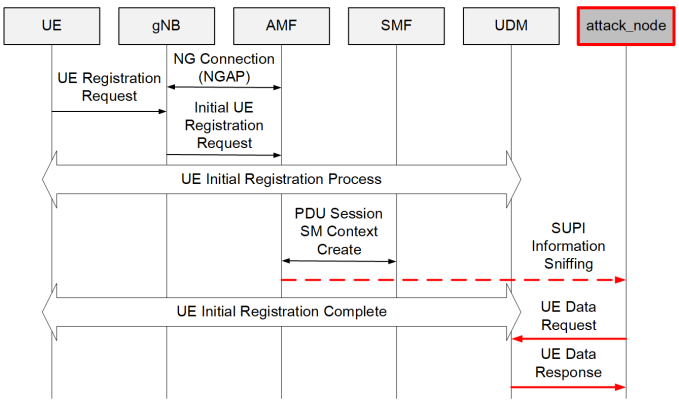


Fig. 9. UDM User Data Hijacking Attack Scenario

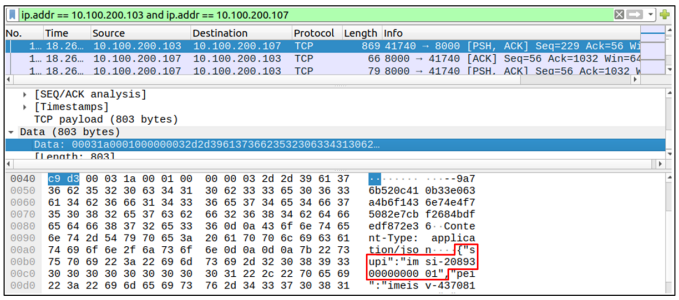


Fig. 10. SUPI Information Sniffing Packet

1) Attack Scenario: The AMF sends a request to the SMF for the UE's SUPI during the initial registration process of the UE to the core network. When the SMF responds, the information is used in the registration process. During this process, an attack_node connected to the network has the ability to intercept this message containing the SUPI information. Once the UE registration process is complete, the attack_node sends a request for UE data information to the UDM using the SUPI information obtained from the intercepted message. Because the UDM lacks an authentication procedure, it responds to the request by sending the UE data to the attack_node without verifying the request's authenticity.

2) Attack Simulation: Fig. 10 depicts the sniffed packet containing the SUPI information of the UE that can be intercepted through packet sniffing. An attacker can analyze the data of this packet and obtain the UE's SUPI information, such as `imsi-208930000000001`, during the UE registration process.

After the UE initial registration process is completed, the SUPI information obtained through packet sniffing is used to send a UE user information request message to the UDM. This is achieved by using a CURL command-based HTTP GET request from the attack_node. The used command is as follows:

```
$ docker exec attack_node curl "http://10.
100.200.108:8000/nudm-sdm/v1/imsi-20893000
0000001/am-data?plmn-id=%7B%22mcc%22%3A%22
208%22%2C%22mnc%22%3A%2293%22%7D"
```

The `docker exec` command is used to execute the curl command within the attack_node container. In this command, the UE data of the SUPI information with the IMSI `imsi-208930000000001` is requested from the UDM using the curl command. The `plmn-id` parameter sets the MCC and MNC of the UE to the information contained within the SUPI value, using URL encoding of `{"mcc":"208", "mnc":"93"}`. Figs. 11 and 12 depict the attack packets and the UDM logs from the attack. Attacks are carried out from `imsi-208930000000001` to

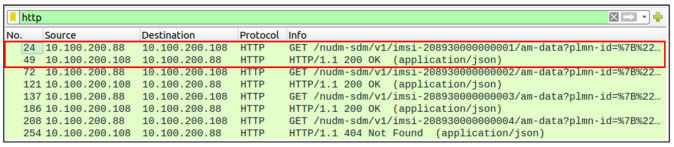


Fig. 11. UDM User Data Hijacking Attack Packet

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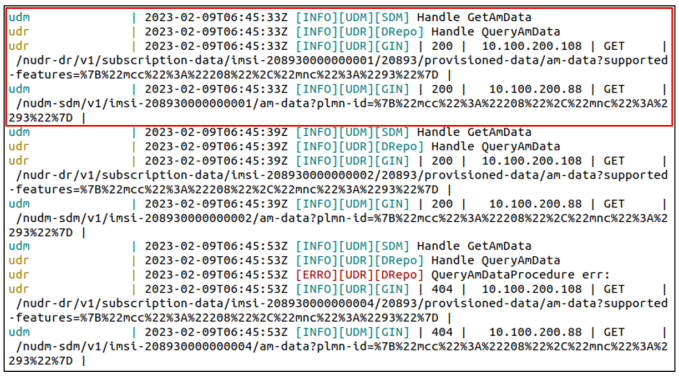


Fig. 12. UDM User Data Hijacking Attack Log

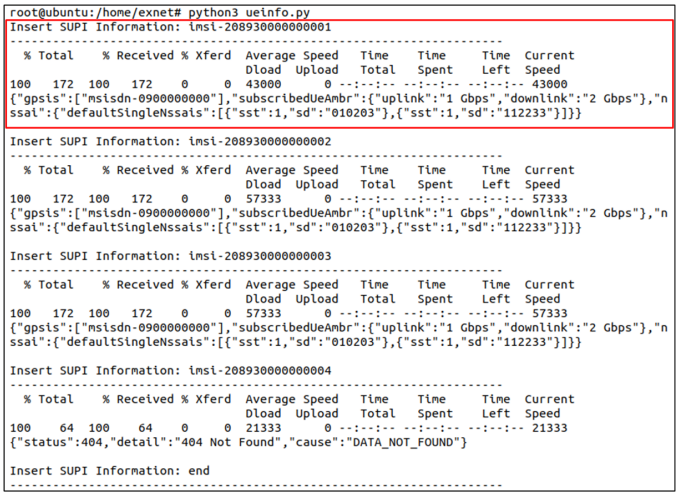


Fig. 13. UDM User Data Hijacking Attack Result

`imsi-208930000000004` using the sniffed SUPI information.

Fig. 13 depicts the outcome of the attack. The UDM supplies the attacker with network slicing information, uplink and downlink speeds, and UE AM subscription information by submitting a request for the UE user information without an authentication mechanism. In the case of `imsi-208930000000004`, the UE identifier was not registered in the UDM, and a 404 Not Found response was returned.

IV. Discussion

It has been assumed that an attack_node has already been connected to a 5G-Advanced network to carry out attacks on its internal components. In the case of an attack_node, it refers to a node installed for experimental convenience, and any network function can potentially become an attack node. If an attacker approaches a vulnerable network function for management reasons, there is a possibility that the attack described in this paper may be carried out in another network that provides services. Therefore, the process of connecting the attack_node to the network is omitted in the experiment. Furthermore, the experiments were conducted in the simulation environment that have been implemented as open-source and not on actual commercial 5G-Advanced networks.

The attack simulation performed in this paper demonstrates the vulnerability of the 5G-Advanced network to data hijacking attacks, due to the lack of encryption and authentication in the internal communication process. This emphasizes the importance of Security by Design. In the 6G era, openness is considered important, and both the core network and RAN are expected to be privatized based on open platforms, so the reliability of the network will be lower than now. 5G-Advanced service providers should therefore adopt appropriate encryption and authentication mechanisms within the core network that it has not considered so far to protect UE data from unauthorized access. It is critical for academics working on the standardization and development of 5G-Advanced networks to achieve a balance between performance and security. The emphasis should be on enhancing both areas, as network security is equally as vital as network performance.

V. Conclusion

In this paper, we have conducted two different data hijacking attacks that take advantage of vulnerabilities present in 5G-Advanced networks where internal communication is not encrypted and internal components are not authenticated. Through our experiment, we demonstrated how an attacker can hijack sensitive information contained within network functions. These findings highlight the importance of implementing robust security measures in the 5G-Advanced core network to prevent such attacks.

Acknowledgment

This work was supported by Institute of Information & communications Technology Planning & Evaluation (IITP) grant funded by the Korea government (MSIT) (No. 2021-0-00796, Research on Foundational Technologies for 6G Autonomous Security-by-Design to Guarantee Constant Quality of Security).

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